

1-Page Performance Cheat Sheet (Meeting-Ready)

System Framing

- Lobby = real-time orchestrator on top of a 3D shell + apps + updates.
- Most issues come from transitions and coupling, not isolated slow code.

Critical User Paths (Golden Paths)

- Boot → Home ready
- Home → App launch
- App → Home return

What We Measure

- Latency: time-to-first-frame, launch duration
- Frame health: jank, long frames, compositor stalls
- Resources: CPU, GPU, memory, IO spikes
- Reliability: crashes, ANRs, launch failures

Likely Systemic Bottlenecks

- Main-thread blocking in lobby/UI
- 3D rendering vs 2D UI contention
- Lifecycle instability during transitions
- Update/version mismatches from Play Store

High-ROI Fix Strategy

- Isolate 3D shell and lobby processes
- Transactional app launches with timeouts/fallbacks
- Crash-loop detection and quarantine
- Update compatibility gating

Optimization Priority

- Tier 1: structural & lifecycle fixes
- Tier 2: IPC, surface, orchestration tuning
- Tier 3: GPU, assets, memory tuning

End Goal

- Fast, deterministic, and update-resilient home experience
- Predictable performance instead of reactive debugging

